1. **Question 1. Descriptive Overview**

Variable	Min	Mean	Max
Opening Gross	4,120,497	17,468,466	68,033,544
Total U.S. Gross	13,090,630	59,620,651	198,000,317
Total Non-U.S. Gross	0	59,560,983	456,235,122
Opening Theatres	852	2,766.28	3,964

Additional Counts

- Number of comedy movies: **23**
- Number of R-rated movies: **15**

2. **Question 2. ROI and Statistical Inference**

The return on investment (ROI) was calculated as the difference between total U.S. box-office gross and budget, divided by the budget. The average ROI across all movies is approximately 29.29%, indicating that, on average, movies tend to generate returns above their production costs.

A 95% confidence interval was constructed for the mean ROI. The resulting interval is [0.1348, 0.4510], suggesting that the true average ROI lies between 13.48% and 45.10%. This indicates that the mean return is consistently positive and economically meaningful.

To evaluate whether the mean ROI exceeds the 12% benchmark cited in the industry, a one-sided hypothesis test was conducted. The test resulted in a p-value of 0.01625, which is below the 5% significance level. Therefore, we reject the null hypothesis and conclude that there is statistically significant evidence that the average ROI is greater than 12%.

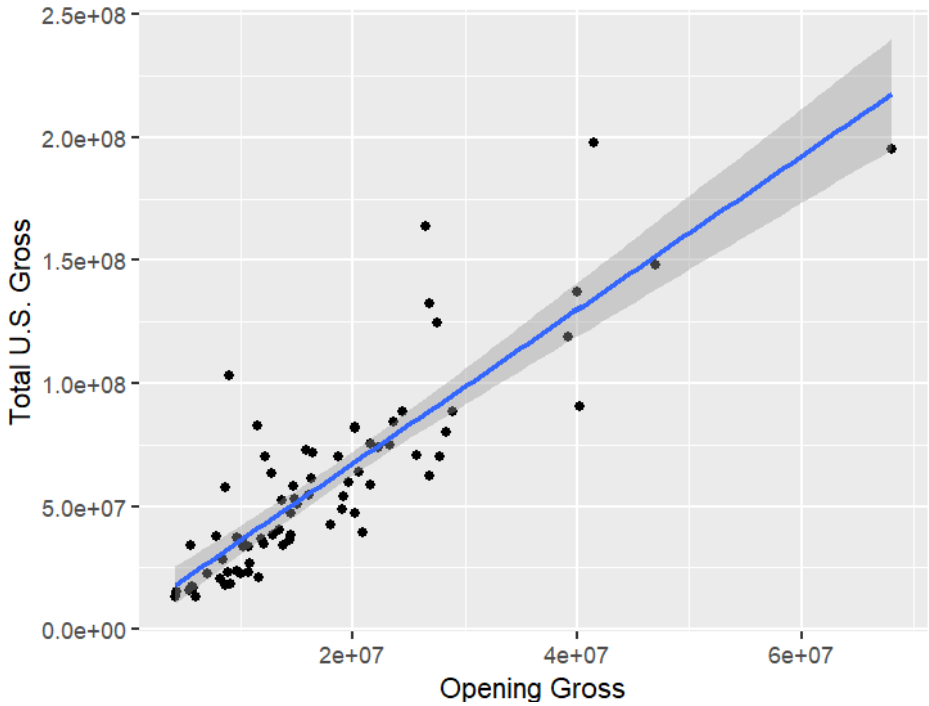
However, it is important to note that the ROI distribution exhibits considerable variability, including negative values, which reflects the high-risk nature of investments in the film industry.

3. **Question 3. Comparison by Genre (Comedy vs Non-Comedy)**

A two-sample t-test was conducted to compare total U.S. gross revenue between comedy and non-comedy movies. The average total U.S. gross for comedy movies is approximately $68.74 million, while for non-comedy movies it is approximately $55.59 million. However, the p-value of 0.1763 indicates that this difference is not statistically significant at the 5% level. Therefore, there is no strong evidence to conclude that comedy movies generate higher box office revenue than non-comedy movies.

A second t-test was performed to compare ROI between comedy and non-comedy movies. The results show that comedy movies have a higher average ROI (54.02%) compared to non-comedy movies (18.36%). The p-value of 0.04743 is below the 5% significance level, indicating that this difference is statistically significant. Therefore, there is evidence that comedy movies tend to generate higher returns on investment than non-comedy movies.

These findings suggest that while comedies do not necessarily produce higher total revenues, they tend to be more efficient in generating returns relative to their budgets.



4. **Question 4. Comparison by MPAA Rating (R-rated vs Non R-rated)**

A two-sample t-test was conducted to compare total U.S. gross revenue between R-rated and non R-rated movies. The average total U.S. gross for non R-rated movies is approximately $61.19 million, while for R-rated movies it is approximately $53.33 million.

However, the p-value of 0.3979 indicates that this difference is not statistically significant at the 5% level. Therefore, there is no evidence to suggest that MPAA rating significantly affects total U.S. box office revenue.

5. **Question 5. Regression Analysis**

a)

A multiple linear regression model was estimated to predict total U.S. box office gross using only factors known before production: budget, genre, MPAA rating, whether the story was previously known, and sequel status. This specification is appropriate because all of these variables can be observed or decided before the film is released.

b)

After estimating the full model, variables that were not statistically significant at the 10% level were removed, based on their p-values exceeding the 10% significance level in the full model. The final reduced model includes Budget, Genre, and Sequel.

The results indicate that budget is a strong positive predictor of total U.S. gross, while sequel status also contributes positively to revenue. Some genre categories also show meaningful differences relative to the base genre. In particular, the sequel variable is statistically significant at the 5% level, indicating that franchise continuation plays an important role in box office performance.

Variable	Coefficient	p-value
Budget	0.8705	<0.001
Sequel	30,252,380	0.0328
Comedy	36,173,610	0.0118
Drama	25,773,250	0.0762

c)

Holding all other explanatory variables constant, sequel movies are expected to have statistically higher total U.S. box office gross than non-sequel movies if the coefficient on the Sequel variable is positive.

In the reduced regression model, the coefficient on Sequel is approximately $30.25 million and is statistically significant at the 5% level (p-value = 0.0328). This indicates

that, controlling for budget and genre, sequel films are expected to generate about $30.25 million more in total U.S. box office revenue than non-sequel movies.

Therefore, there is statistical evidence that sequel movies outperform non-sequels in terms of total U.S. gross, holding other factors constant. This result provides evidence that franchise continuation plays a statistically significant role in box office performance.

6. **Question 6. Opening Gross Regression Analysis**

a)

A multiple linear regression model was estimated to predict opening weekend box office gross. The model includes both preproduction variables (budget, genre, MPAA rating, known story, and sequel status) as well as variables related to the opening weekend (number of opening theatres and seasonal release indicators such as summer, holiday, and Christmas).

This specification is appropriate because it incorporates all relevant factors that are known before or at the time of release and that may influence initial box office performance. This allows us to isolate the marginal effect of each variable on opening gross revenue.

b)

After estimating the full regression model, all variables that were not statistically significant at the 10% level were removed, based on their individual p-values. The final reduced model includes Budget, MPAA rating, Sequel, Opening Theatres, and Summer.

All variables included in the reduced model are statistically significant at the 10% level, based on their individual p-values. The results indicate that Opening Theatres is the most important predictor of opening gross. Budget and sequel status are also statistically significant and positively related to opening performance. Additionally, MPAA rating and seasonal timing contribute to explaining variation in opening weekend revenue.

Variable	**Coefficient**	**p-value**	**Interpretation**
Opening Theatres	7,899	<0.001	+$7,899 per theatre
Budget	0.151	<0.001	Positive effect
Sequel	9.36M	0.0038	Higher than non-sequel

Variable	Coefficient	p-value	Interpretation
MPAA PG	9.80M	0.054	Marginally significant
MPAA PG-13	12.69M	0.012	Higher than base
MPAA R	10.68M	0.039	Higher than base
Summer	-4.21M	0.044	Negative effect

c)

The estimated coefficients from the reduced regression model can be interpreted as follows, holding all other variables constant:

- **Opening Theatres:** Each additional theatre in which a movie is shown during its opening weekend increases expected opening box office revenue by approximately $7,899. This highlights the critical role of distribution scale in determining initial performance.
- **Budget:** The coefficient on Budget is positive and statistically significant, indicating that higher production and marketing investment leads to higher opening weekend revenue. Specifically, an increase of one dollar in budget is associated with an increase of approximately $0.151 in opening gross, ceteris paribus.
- **Sequel:** The coefficient on Sequel is positive and statistically significant, indicating that sequel movies are expected to generate approximately $9.36 **million more** in opening weekend revenue than non-sequels, holding all other variables constant. This reflects the advantage of established audiences and brand recognition.
- **MPAA Rating:** The coefficients associated with MPAA rating categories represent differences in opening revenue relative to the omitted base category (likely G or NR). Movies rated PG, PG-13, and R show higher opening gross compared to this reference group, indicating that audience targeting and content rating play a role in initial box office performance.
- **Summer:** The coefficient on the Summer variable is negative and statistically significant, indicating that, controlling for all other factors, movies released during the summer generate approximately $4.21 million less in opening weekend revenue, possibly due to increased competition during peak release periods.

Overall, the results suggest that opening weekend performance is driven not only by investment and distribution scale, but also by positioning and audience segmentation.

d)
Based on the reduced regression model, the estimated coefficient for the number of opening theatres is approximately 7,899. Therefore, holding all other variables constant, an increase of 100 theatres is associated with an expected increase of approximately $790,000 in opening weekend box office revenue.

Using the 95% confidence interval for the slope coefficient, the corresponding confidence interval for the effect of 100 additional theatres is approximately [$499,907,\ $1,079,956].

This result highlights the strong and statistically significant impact of distribution scale on opening weekend performance.

7. **Question 7. Relationship Between Opening Gross and Total U.S. Gross**

a)

A simple linear regression was estimated to analyze the relationship between opening weekend gross and total U.S. box office gross. The results show that opening gross is a highly significant predictor of total revenue, with a coefficient of approximately 3.12 and a p-value below 0.001.

b)

If the industry “25% rule” were true, meaning that opening weekend revenue represents 25% of total U.S. gross, then total gross would be four times the opening gross. Therefore, the slope coefficient in the regression would be equal to 4.

c)

To formally evaluate the industry “25% rule,” we test the null hypothesis $H_0: \beta_1 = 4$against the alternative $H_1: \beta_1 \neq 4$, where $\beta_1$is the slope coefficient from the regression of total U.S. gross on opening weekend gross.

The estimated slope coefficient is approximately 3.12. A formal t-test yields a test statistic of -4.03 with a p-value of 0.00013. Additionally, the 95% confidence interval for the slope is [2.686, 3.555], which does not include the value 4 implied by the 25% rule.

Therefore, we reject the null hypothesis at the 5% significance level and conclude that the 25% rule does not hold for this dataset. In particular, opening weekend revenue represents a larger share of total revenue than the traditional benchmark suggests.

Parameter	Value
Estimated slope (β_1)	3.12
t-statistic (H_0: $\beta_1 = 4$)	-4.03
p-value	0.00013
95% Confidence Interval	[2.686, 3.555]

d)

Although a formal hypothesis test was conducted in part (c), the analysis based on the simple linear regression has important limitations. In particular, the model may suffer from omitted variable bias, as it does not account for other relevant factors such as budget, genre, MPAA rating, or sequel status, which can also influence total box office performance.

As a result, the estimated relationship between opening weekend gross and total U.S. gross may be capturing the effects of these omitted variables. Therefore, while the 25% rule is statistically rejected in this simple framework, a more comprehensive model is required to obtain a more accurate and reliable assessment of this relationship.

e)

To improve the analysis, a multiple regression model was estimated including additional explanatory variables such as budget, genre, MPAA rating, and sequel status.

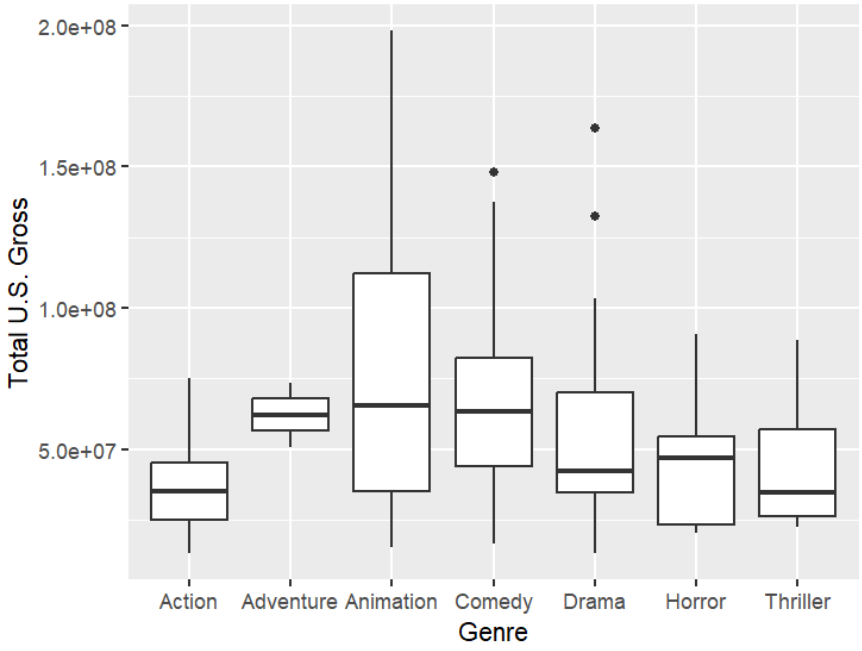
f)

Using the extended regression model, opening gross remains a highly significant predictor of total revenue, with a coefficient of approximately 3.06 and a p-value below 0.001. This confirms that opening performance continues to be a key determinant of total box office success, even after controlling for other relevant variables.

In addition, budget is also statistically significant and positively related to total revenue, indicating that higher investment levels contribute to stronger overall performance. Some genre effects, particularly drama, also appear to be statistically significant, while other variables such as MPAA rating and sequel status are not significant in this specification.

g)

The extended model shows an R-squared of approximately 0.833, indicating that about 83.3% of the variation in total U.S. box office gross is explained by the variables included in the model. This represents a substantial improvement compared to the simple regression, highlighting the importance of including additional explanatory variables to better capture the determinants of box office performance.



8. **Question 8. Impact of Critics' Opinion on Box Office Performance**

a)

A multiple linear regression model was estimated to analyze the impact of critics' opinion on total U.S. box office gross, controlling for opening gross, budget, genre, MPAA rating, and sequel status.

The results show that critics' opinion is a statistically significant predictor of total revenue, with a positive coefficient of approximately 572,300 and a p-value of 0.00143. This indicates that higher critical ratings are associated with higher box office performance, even after controlling for other important factors.

As expected, opening gross remains the strongest predictor of total revenue, with a highly significant coefficient of approximately 2.86 and a p-value below 0.001. This confirms that early performance continues to be a key determinant of overall success.

The budget variable also remains statistically significant and positively related to total revenue, reinforcing the importance of investment scale. In addition, some genre effects, particularly comedy and drama, show statistically significant positive contributions to total gross.

In contrast, MPAA rating and sequel status are not statistically significant in this specification, suggesting that their effects may be less relevant once opening performance and critics' opinion are taken into account.

The model exhibits a strong explanatory power, with an R-squared of approximately 0.859, indicating that about 85.9% of the variation in total U.S. gross is explained by the variables included.

Overall, the results suggest that critics' opinion plays a meaningful role in explaining box office performance, complementing the strong predictive power of opening weekend revenue.

b)

After estimating the full regression model, variables that were not statistically significant at the 10% level were removed. The final model retains Opening Gross, Critics' Opinion, Budget, and Genre as explanatory variables.

The results show that Opening Gross is the strongest predictor of total U.S. box office revenue, with a highly significant and positive coefficient. Critics' opinion is also statistically significant and positively related to total revenue, indicating that better reviews contribute to higher box office performance.

Budget remains statistically significant, suggesting that higher investment levels are associated with increased revenue. Regarding genre, comedy and drama films show statistically significant positive effects relative to the base category, while other genres are not significant.

The model exhibits strong explanatory power, with an R-squared of approximately 0.853, indicating that about 85.3% of the variation in total U.S. gross is explained by the included variables.

Variable	**Coefficient**	**p-value**
Opening Gross	2.76	<0.001
Critics' Opinion	542,400	<0.001
Budget	0.291	0.0049
Comedy	19.23M	0.0084
Drama	18.34M	0.0131

c)

Using the reduced regression model, a prediction was constructed for a movie with characteristics similar to Flags of Our Fathers. Based on the model, the expected total U.S. gross revenue is approximately $98,734,372.

The corresponding 95% prediction interval is [$63,890,631,\ $133,578,113], reflecting the uncertainty associated with predicting the performance of an individual movie.

Although the point estimate suggests strong box office performance, the prediction interval is relatively wide, which highlights the high level of uncertainty and risk inherent in movie investments, even when important explanatory variables are taken into account.

Estimate	Lower 95%	Upper 95%
$98,734,372	$63,890,631	$133,578,113

9. **Question 9. Interaction Between Critics' Opinion and Comedy Genre**

To examine Griffith's claim, the regression model from Question 8 was modified by including an interaction term between critics' opinion and a dummy variable for comedy films. This allows the effect of critics' opinion on total U.S. box office gross to differ between comedies and non-comedies.

The results show that the coefficient for critics' opinion is positive but not statistically significant (p-value = 0.157), indicating that critics' ratings do not have a strong standalone effect on total revenue once other factors are controlled for.

More importantly, the interaction term between critics' opinion and comedy is also not statistically significant (p-value = 0.380). This suggests that the effect of critics' opinion on box office performance is not significantly different for comedies compared to non-comedies.

Therefore, there is no statistical evidence to support Griffith's claim that critics' opinion has a smaller influence on total U.S. gross for comedies than for other genres.

10. **Question 10. Star Power vs Budget**

If a variable measuring star power, defined as the number of A-list actors in a movie, were added to the regression model from Question 9, Griffith's conclusion would be supported only under two conditions.

First, the slope coefficient of the star power variable would need to be positive and statistically significant, indicating that movies with more A-list stars tend to generate higher total U.S. box office gross, holding all other variables constant.

Second, after including star power in the model, the slope coefficient of the budget variable would need to decrease substantially in magnitude or become statistically insignificant. This would suggest that the positive effect previously attributed to budget was actually capturing, at least in part, the influence of star actors embedded in larger-budget productions.

Therefore, Griffith's interpretation would be correct only if star power enters the regression as a significant positive predictor and the independent explanatory effect of budget weakens once star power is controlled for.